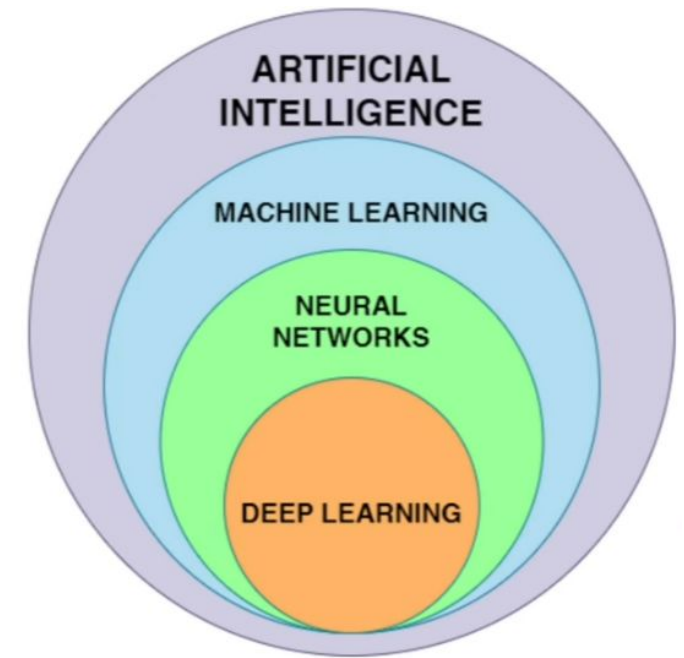


AI Hackathon - Design Change Identification

Object detection using YOLOv8 (layman's perspective)

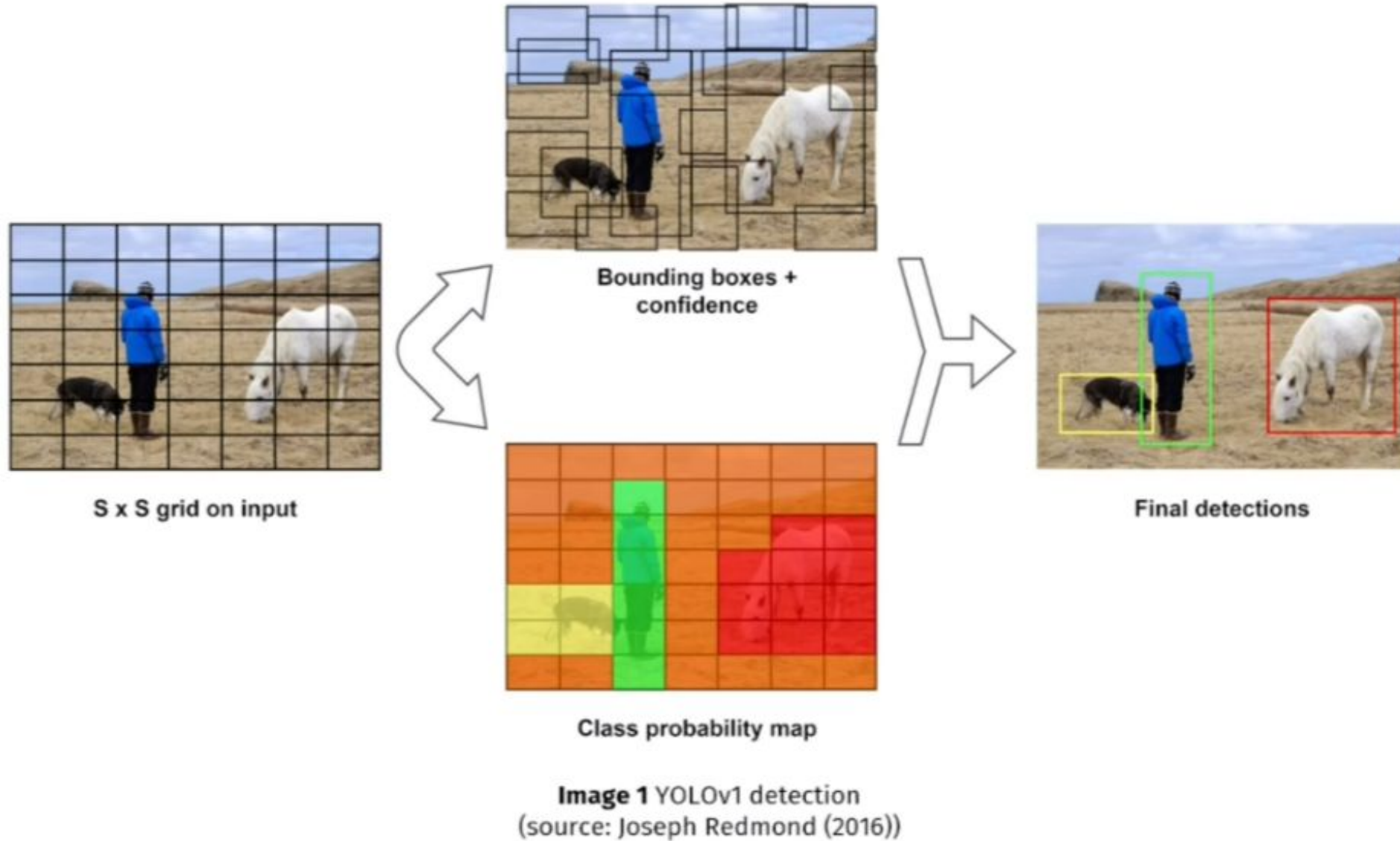
YOLOv8 is an object detection model that uses **convolutional neural network (CNN)** to detect and predict objects from an image.



Subsets of AI



AI Hackathon - Design Change Identification



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Step 1. Getting dataset

Available datasets from [Kaggle.com](https://www.kaggle.com)



+ Create

Home

Competitions

Datasets

Models

Code

Discussions

Learn

More

Search

WindowDetectionFloorPlans

1

New Notebook

Data Card

Code (0)

Discussion (0)

Suggestions (0)

test (14 files)

002eb862-2023-12-06...
166.46 kB

00ceb44f-2023-12-06...
12.56 kB

05ad98dd-2023-12-06...
15.7 kB

1deaa44c-2023-12-06...
90.53 kB

3bfed6a3-2023-12-06...
17.5 kB

3e59ddc7-2023-12-06...
18.41 kB

4ac32531-2023-12-06...
314.18 kB

4dba58ba-2023-12-06...
158.11 kB

5b156dfb-2023-12-06...
226.53 kB

5d82ac3e-2023-12-06...
188.9 kB

7e2af234-2023-12-06...
208.1 kB

7fa2b2bd-2023-12-06...
22.44 kB

8d7b6194-2023-12-06...
118.09 kB

9f1a419a-2023-12-06...
17.21 kB

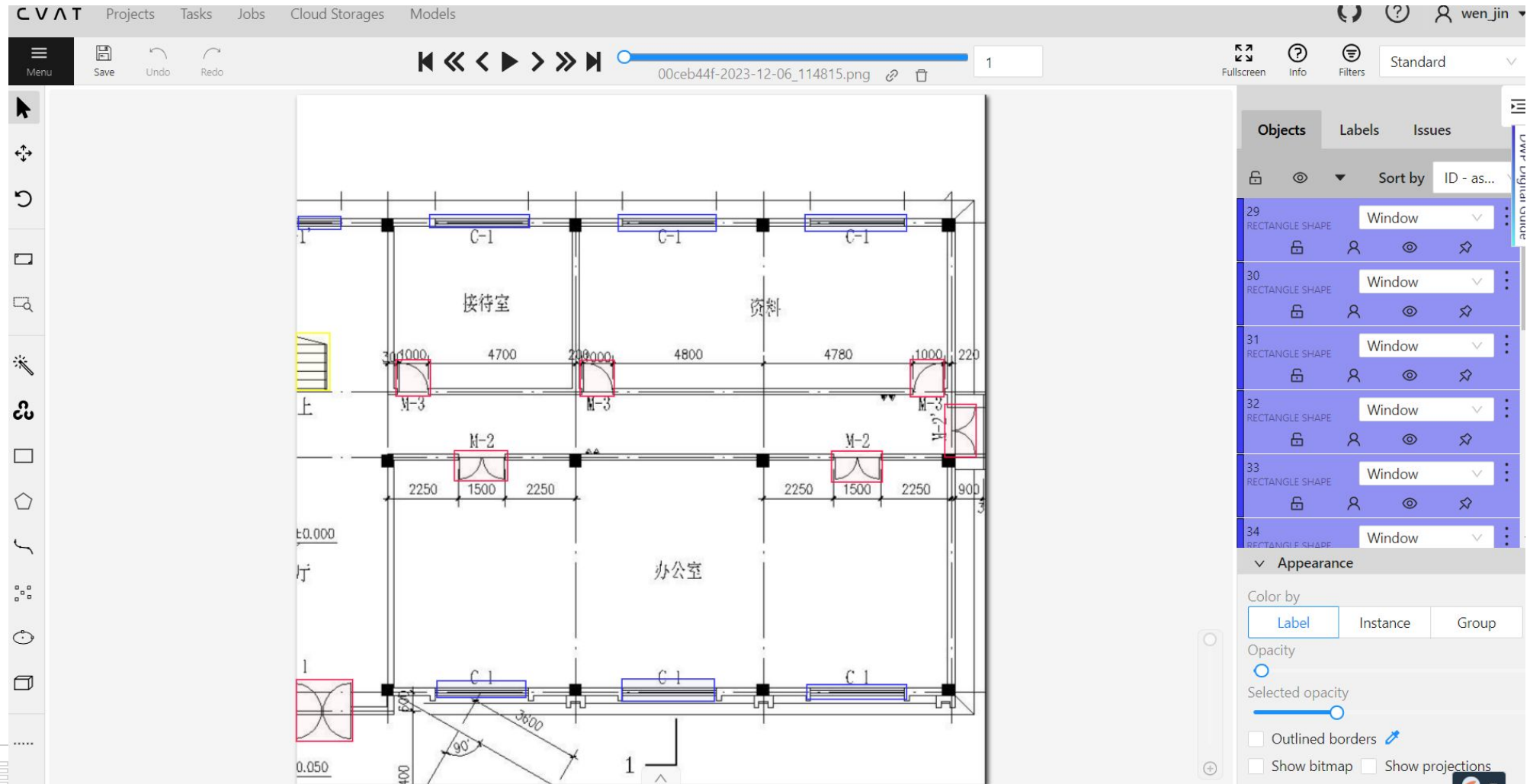
3

AI Hackathon - Design Change Identification

Step 2. Labelling

Label data on
www.cvat.ai

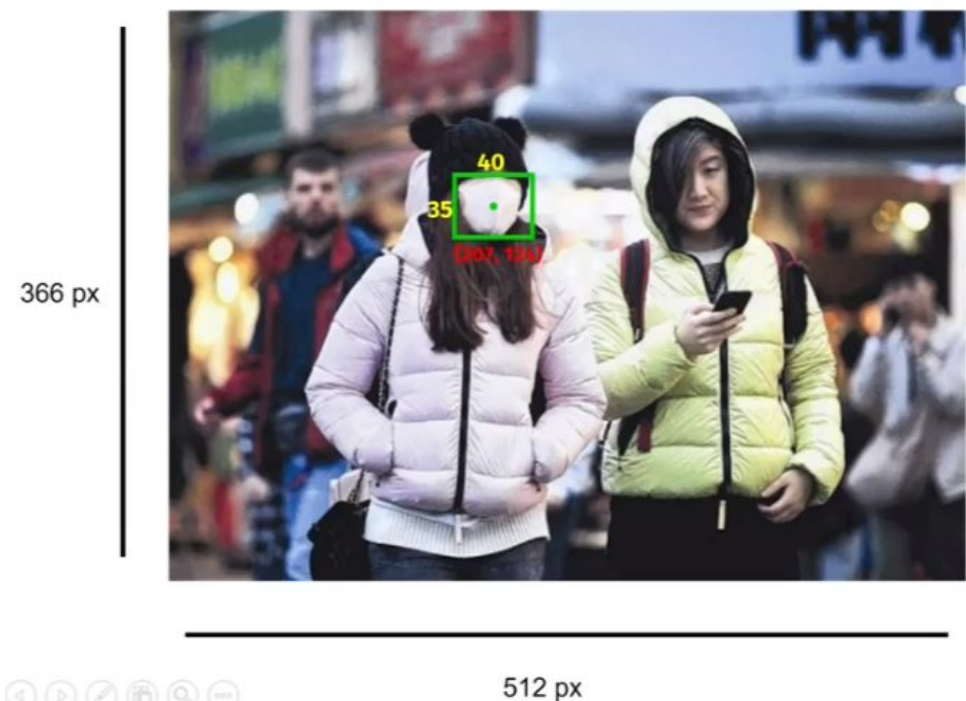
Labeled objects:
Doors
Windows



AI Hackathon - Design Change Identification

Step 3. Export dataset

Convert the labeled dataset into specific format required for training the model.



```
<object-class> <x> <y> <width> <height>

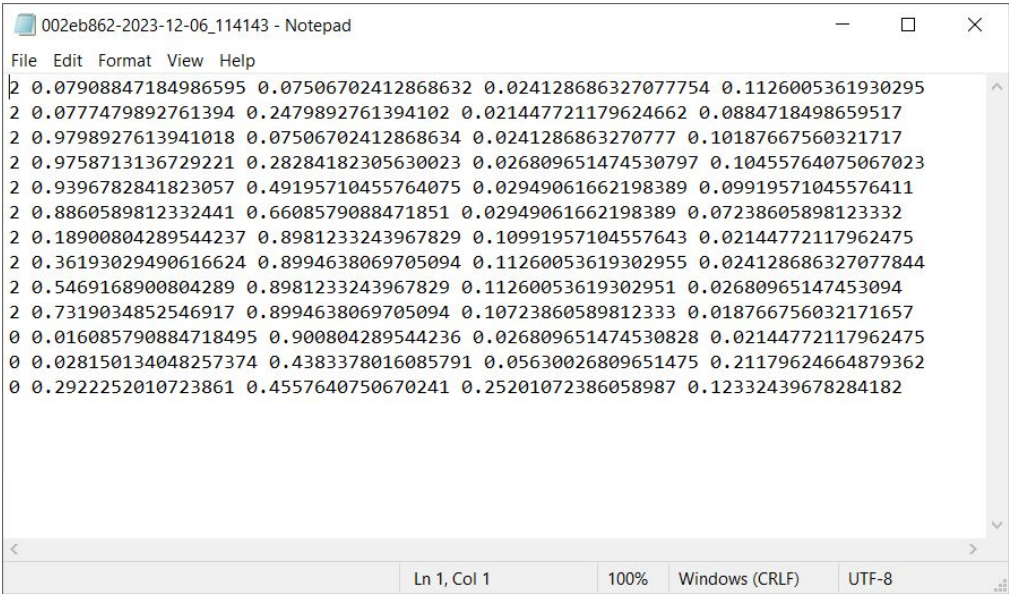
0 0,404297 0,338798 0,078125 0,095628
```

$x = 207 / 512 = 0,404297$

$y = 124 / 366 = 0,338798$

$width = 40 / 512 = 0,078125$

$height = 35 / 366 = 0,095628$



Exported .txt file



AI Hackathon - Design Change Identification

Step 4. Running scripts

Training the model on
Google Colab

Output: **best.pt** file

Trained objects:
Doors
Windows

YOLOv8.ipynb

File Edit View Insert Runtime Tools Help Last edited on March 23

+ Code + Text

Training Process

```
[ ] # Train
!yolo detect train model=yolov8l.pt data=data/face_mask.yaml imgsz=640 workers=8 batch=8 device=0 epochs=300 patience=50 name=yolov8_face_mask
```

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	4.97it/s]
	all	6	133	0.697	0.617	0.666	0.425				

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size					
103/300	6.37G	1.083	0.7073	1	24	640: 100% 7/7	[00:02<00:00,	2.83it/s]			
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	5.45it/s]
	all	6	133	0.527	0.704	0.653	0.385				

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size					
104/300	6.34G	1.082	0.7847	1.014	10	640: 100% 7/7	[00:02<00:00,	2.71it/s]			
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	3.47it/s]
	all	6	133	0.627	0.536	0.602	0.36				

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size					
105/300	6.34G	1.123	0.7526	0.998	49	640: 100% 7/7	[00:02<00:00,	2.85it/s]			
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	6.19it/s]
	all	6	133	0.682	0.566	0.583	0.335				

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size					
106/300	6.43G	1.147	0.7547	1.021	24	640: 100% 7/7	[00:02<00:00,	2.57it/s]			
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	3.16it/s]
	all	6	133	0.514	0.756	0.687	0.411				

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size					
107/300	6.36G	1.108	0.734	1.005	33	640: 100% 7/7	[00:02<00:00,	2.83it/s]			
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	6.70it/s]
	all	6	133	0.747	0.777	0.755	0.459				

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size					
108/300	6.41G	1.032	0.6812	0.9945	30	640: 100% 7/7	[00:02<00:00,	2.75it/s]			
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	6.09it/s]
	all	6	133	0.753	0.782	0.755	0.472				

epochs

accuracy



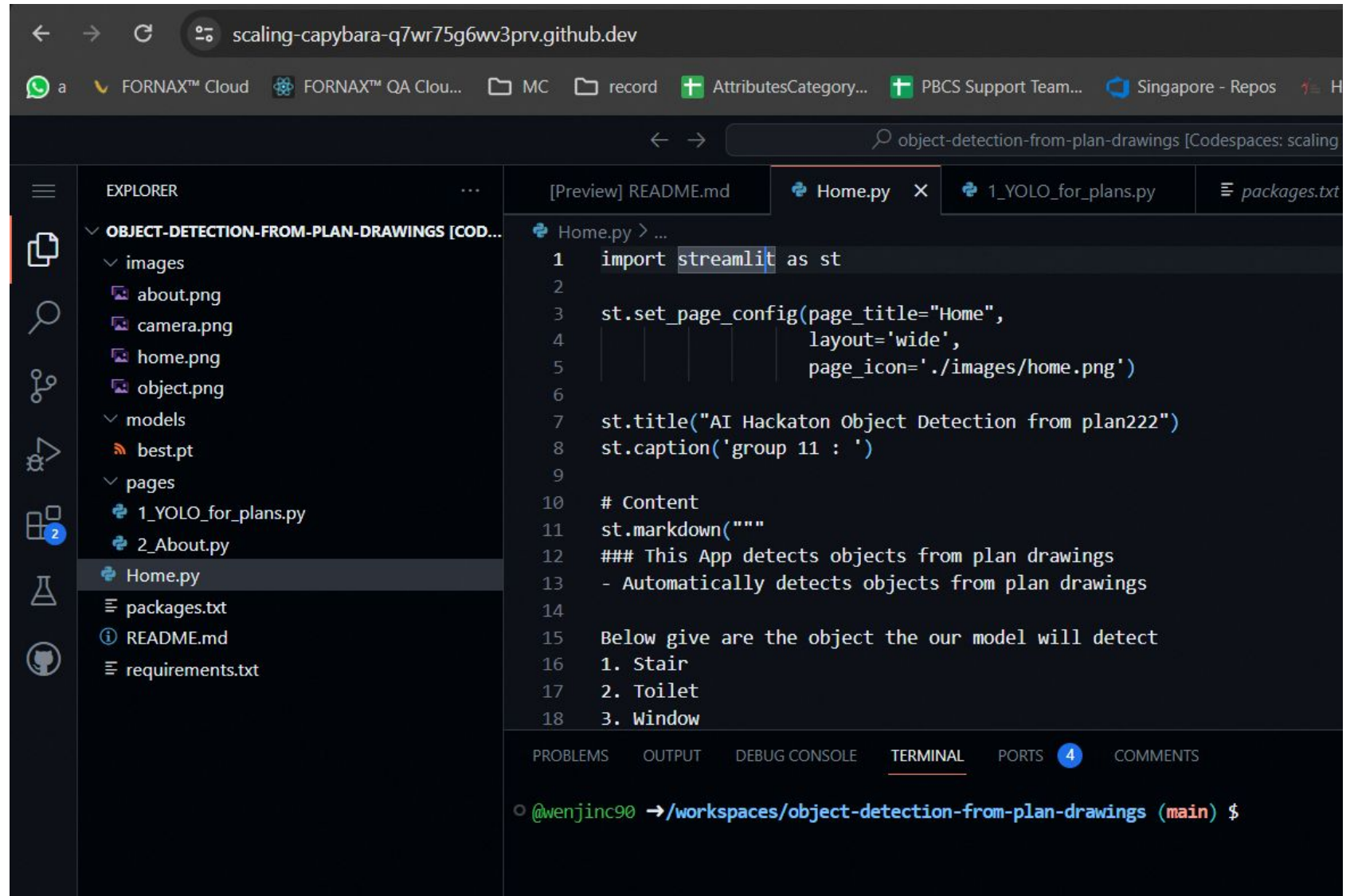
AI Hackathon - Design Change Identification

Step 5. Publishing

Writing the webpage
on [GitHub](#) then hosting
the app on
[Streamlit.com](#)

Learning platforms:

- [LEARN](#)
([Learn.gov.sg](#))
- [Microsoft Copilot](#)



The screenshot shows a GitHub Codespace environment for a project named "object-detection-from-plan-drawings". The Explorer panel on the left shows the file structure:

- OBJECT-DETECTION-FROM-PLAN-DRAWINGS [COD...]
 - images
 - about.png
 - camera.png
 - home.png
 - object.png
 - models
 - best.pt
 - pages
 - 1_YOLO_for_plans.py
 - 2_About.py
 - Home.py (selected)
 - packages.txt
 - README.md
 - requirements.txt

The main editor shows the content of `Home.py`:

```
1 import streamlit as st
2
3 st.set_page_config(page_title="Home",
4                     layout='wide',
5                     page_icon='./images/home.png')
6
7 st.title("AI Hackaton Object Detection from plan222")
8 st.caption('group 11 : ')
9
10 # Content
11 st.markdown("""
12 ### This App detects objects from plan drawings
13 - Automatically detects objects from plan drawings
14
15 Below give are the object the our model will detect
16 1. Stair
17 2. Toilet
18 3. Window
```

The bottom panel shows the TERMINAL with the command prompt:

```
@wenjinc90 →/workspaces/object-detection-from-plan-drawings (main) $
```



AI Hackathon - Design Change Identification

1. Scalability and Impact for Architects and Construction

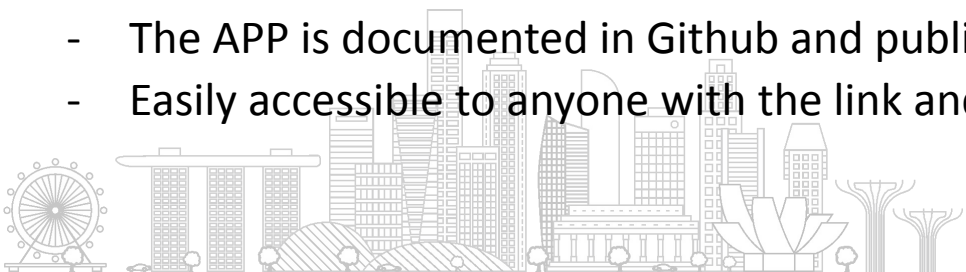
- Cutting down time and effort to meticulously examine each drawing, pinpoint variances and document the changes for each RVO and costing purpose.
- Auto detection and identification of changes in drawings submitted by consultants to contractors. Similar to how consultants need to produce revision cloud.
- Scale up to automate the preparation drawings for amendments to approved plan submission, Recon and A&A projects. (Magenta, Cyan, Yellow)

2. Innovation and Integration in Architecture and Construction

- Convolutional Neural Network (CNN) is widely used for facial recognition, medical imaging, autonomous vehicle, but not much actual use cases for the construction industry.
- Automating the manual task of identifying changes on drawings greatly reduce risk of missing out certain changes made to the design.
- Embedded into file exchange platforms as a pre-checker to quickly notify the sender if the drawings are in order.

3. Technical Documentation and Execution in Architectural Projects

- The APP is documented in Github and published on Streamlit.
- Easily accessible to anyone with the link and internet is free to test it out.



Thank you